

DEPARTEMENT OF COMPUTER ENGINEERING

Academic Project Proposal

Project Topic	PharmTwinAI – AI-Based Digital Twin for Smart Pharmacy Inventory, Demand Forecasting, and Decision Support
Organization	Bhagyashree Medical (<i>Industry Collaboration</i>) Juinagar, Navi Mumbai, Maharashtra, India. Point of Contact (POC): Sandhya Ghemud
Project Team Members	Atharva Bendale, Neha Ghemud, Sachi Gupta, Ishan Mishra
Abstract	PharmTwinAI is an AI-powered Digital Twin platform designed to digitally replicate pharmacy operations for real-time inventory monitoring and intelligent decision support. The project addresses challenges such as medicine stock shortages, overstocking, expired medicines, and inaccurate demand forecasting. The proposed solution involves collecting historical inventory and sales data, creating a virtual pharmacy model, implementing machine learning algorithms for demand prediction, and developing an interactive dashboard for monitoring stock levels and generating recommendations. The expected outcome is an intelligent software system capable of improving inventory utilization, reducing medicine wastage, enhancing operational efficiency, and supporting better business decisions. The project demonstrates the practical application of Artificial Intelligence and Digital Twin technology in the healthcare retail sector.
Deliverables	Deliverables • AI-based Digital Twin software for pharmacy operations. • Interactive web dashboard for inventory monitoring. • Medicine demand forecasting module. • Low-stock and expiry alert system. • Inventory analytics and reporting dashboard. • Project documentation and user manual.

	• Final project demonstration. Beneficiaries: • Bhagyashree Medical • Pharmacy staff and management • Customers through improved medicine availability • Academic institution for industry-based research
Required facilities (Necessary resources) :	• Computer Laboratory • Internet Connectivity • Python Development Environment • VS Code • MySQL Database • Power BI / Streamlit Dashboard • Cloud Storage • Pharmacy inventory dataset provided by Bhagyashree Medical
Knowledge areas needed for project	• Artificial Intelligence • Machine Learning • Digital Twin Technology • Python Programming • Data Analytics • Database Management (MySQL) • Cloud Computing (basic) • Web Development (Flask/Streamlit) • Data Visualization (Power BI)
Proprietary information and Confidentiality requirement	i) Is the confidentiality required for information shared with the students or not. Yes. Confidentiality is required for all pharmacy inventory, sales, customer, and operational data shared with the student team. The data will be used strictly for academic and project development purposes ii) Intellectual Property owner rights retained by sponsor / students /both. Both. Intellectual Property (IP) rights will be jointly held by the sponsor (Bhagyashree Medical) and the student project team, subject to the institute's project policies and mutual agreement.

	iii) Sponsor accepts responsibility to discuss IP ownership directly with the student's projects team and faculty project advisor / supervisor. Yes. The sponsor agrees to discuss and finalize Intellectual Property ownership, usage rights, and confidentiality terms with the student project team and the faculty project advisor/supervisor before project completion.
Estimated Budget	0
Project Duration	6 Months

Name of the students

Roll Number

Signature

1) Atharva Bendale

123A1017

2) Neha Ghemud

123A1036

3) Sachi Gupta

123A1037

4) Ishan Mishra

123A1040

To be filled by the Department.

This proposed project topic” _____

_____ “ has been reviewed and recommended for consideration. I am ready to accept this project to be guided and will see its progress and completion within the timeline decided by the department. Also, it will be reported and recommended to the authority in the department to disqualify the project if any plagiarism is committed by the students.

Designated Guide / Recommended By

Based on the above recommendation, the project has been accepted by the Department of

_____ in the meeting held on

_____ for assigning it to the requested students during the academic

year _____. The following faculty members have been requested to be the guide

for this student project.

Name of the Project Guide : Prof.

Signature:

Head,

Department of _____

Dept. Project Co-ordinator/s :